

Set 2 - Graphs

Remark: All graphs are considered simple.

1. Construct, when it is possible, a graph having the following degrees sequences. When it is not possible, explain why. $(1, 1, 1, 1, 4)$, $(1, 2, 2, 2, 3)$, $(0, 1, 2, 3, 4)$, $(1, 1, 2, 2, 4)$, $(2, 2, 2, 2, 2)$, $(1, 1, 1, 3, 3)$, $(2, 2, 2, 2, 2, 2)$, $(3, 3, 3, 3)$, $(1, 2, 3, 3, 3)$, $(1, 1, 1, 1, 1, 1)$, $(2, 2, 3, 4, 4, 5)$, $(1, 2, 2, 3, 3, 4)$, $(1, 2, 3, 3, 3, 4, 4)$, $(1, 1, 2, 3, 3, 4, 4, 6)$, $(3, 3, 3, 3, 3, 5, 5, 5)$, $(2, 2, 3, 3, 3, 3, 3, 3, 4, 4)$, $(1, 1, 3, 3, 4, 4, 6, 6)$, $(1, 3, 3, 4, 5, 6, 6)$, $(2, 3, 3, 4, 5, 6, 7)$.
2. Construct, if it is possible, a graph with 13 vertices and 31 edges, with 3 vertices of degree 1 and 7 vertices of degree 4.
3. a) Prove that in a graph with n vertices ($n \geq 3$) which does not contain triangles (i.e. does not contain cycles of length 3) the sum of degrees of every two adjacent vertices is at most 3.
b) In a contest there are 15 participants and each of them must play a game with each other participant. During the contest one remark that, no matter how one chooses three persons, there exist two which haven't play yet together. Show that there exist at least one person which has already played at most 8 games.
c) Which is the maximum number of edges of a graph with n vertices which does not contain triangles? For which kind of graphs this maximum is attained?
d) Construct a connected graph with n vertices ($n \geq 5$) which contains at least a triangle, such that the sum of degrees of every two adjacent vertices is at most n .
4. Let G be a graph with 10 vertices which does not contain triangles. Prove that, for every two adjacent vertices, there exist one of degree at most 5.
5. a) Let G be a graph with at least 6 vertices. Prove that G or $\overline{G}$ contains triangles.
b) Prove that, at each moment of a contest with at least 6 teams, there exist 3 teams which played between them, or 3 teams which haven't played yet any game between them.
6. Let $G = (V, E)$ be a graph with n vertices, with $\deg(v) \geq \frac{n-1}{2}$ for all $v \in V$. Prove that G is connected.
7. We have a genuine coin labeled 0 and n other coins, indistinguishable from 0 by appearance, except from being labeled $1, 2, \dots, n$. It is suspected that one coin is false – either too light or too heavy. Show that at least $\lceil \log_3(2n+1) \rceil$ weighings on a balance are needed to decide which coin (if any) is false, and also to decide whether it is light or heavy. Devise a procedure which uses exactly this number of weighings, when $n = 4$ and $n = 6$.